

HIMANSHU RANJAN

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EDUCATION

MS Ramaiah Institute of Technology

B.E. in Artificial Intelligence and Data Science - **CGPA - 9.4**

2023 - 2027

Bangalore, India

COURSEWORK / SKILLS

- Data Structures & Algorithms
- Operating Systems
- Data communication and Network
- OOPS Concept
- Artificial Intelligence and Machine Learning
- Web Development
- Database Management System (DBMS)
- Data Science

EXPERIENCE

Samsung PRISM

Samsung PRISM Research Intern

Feb 2025 - Jun 2025

Bangalore, India

- Conducted research on **speaker-independent speech emotion recognition** and **emotion-aware wake word systems** using **PyTorch**, **Transformers**, and **Wav2Vec2**.
- Designed a hierarchical **arousal-valence emotion classification pipeline** using wav2vec2, attention pooling, focal loss, acoustic feature fusion, and temporal smoothing techniques.
- Implemented **LOSO evaluation pipelines** to eliminate speaker leakage and improve experimental rigor using **scikit-learn**, **NumPy**, and **Pandas**.
- Integrated eGeMAPS acoustic features using **Librosa**, weak domain-adversarial training, and utterance-level **HMM smoothing**.
- Developed optimized GPU-only mixed precision training workflows using **CUDA**, **Torch AMP**, and **DataLoader**-based batching for efficient large-scale experimentation.

PROJECTS

StockPredictor | Tech Stack Used : Python, Typescript

- Built a responsive **TypeScript** UI with reusable components and interactive stock charts
- Model - Implemented an **LSTM**-based stock predictor with preprocessing, sliding-window training, and MSE/RMSE evaluation.
- Integration - Using **fastapi**

Money-Tracker | Technology Stack Used : Flutter, Hive

- An offline-first Budget tracking app.
- **Biometric authentication** with PIN/Password fallback for app security.
- Implemented **PDF export** with transaction history, payment details, and polite reminder message.
- **Trie** data-structure implementation for the **auto suggestion and categorization** of the budget.

EdgeSync | Technology Stack Used : Dart, Python, TensorFlow

- Developed noise injection methods using **frequency DCT** to poison the stable diffusion model training on private images.
- Built **TFLite** models for mobile-side prediction of image perturbation parameters.
- Achieved SSIM between **0.80-0.88** for the noised images.

Cpu-Scheduling-Algorithms | Technology Stack Used : Python, Node.js, Typescript

- Built a **fullstack** CPU scheduling simulator supporting multi-core execution and scheduling visualization.

TECHNICAL SKILLS

Languages: Python, C, C++, Dart, SQL

Web Development: HTML, CSS, JavaScript

Database Management: MySQL, MongoDB

Developer Tools: VS Code, Android Studio

Technologies/Frameworks: Linux, GitHub, Git, Flutter

Achievements

- Among **Top 10** finalist in the hackathon known as Kodikon, conducted by PES Engineering university.
- Selected for the final round of the **Samsung Prism** Hackathon